

SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE (SMVEC)

DEPARTMENT OF PHYSICS

PROJECT REPORT

ON

**“HEART RATE MONITOR USING ARDUINO
(TINKERCAD)”**

Submitted by

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CERTIFICATE

This is to certify that THEERSHITHA S (Register No. 25UPHY012), a student of B.Sc. Physics, Second Year, III Semester, Sri Manakula Vinayagar Engineering College (SMVEC), has completed the academic project entitled “Heart Rate Monitor Using Arduino (Tinkercad)” as part of the academic requirements of the course.

Project Guide: _____

Head of the Department: _____

DECLARATION

I hereby declare that this project report entitled “Heart Rate Monitor Using Arduino (Tinkercad)” is prepared by me for academic purposes. The work explains the principle of heart-rate sensing, Arduino-based signal processing, circuit connections, simulation, applications, advantages and limitations. The project is intended as an educational prototype and not as a clinical diagnostic device.

Signature of Student: _____

Date: _____

ABSTRACT

Heart rate is an important physiological parameter that indicates how frequently the heart beats in one minute. This project presents the design and simulation of a simple heart rate monitoring system using an Arduino board and a pulse sensor in Tinkercad. The pulse sensor detects periodic changes in blood volume associated with cardiac activity. Its output is converted into an electrical signal and read by the Arduino through an analog input. The Arduino program detects pulse peaks, estimates the time interval between successive beats and calculates beats per minute (BPM). The measured value can be displayed on a serial monitor or an LCD in a practical hardware implementation.

The project demonstrates the interdisciplinary connection between physics, electronics, instrumentation and basic biomedical sensing. Tinkercad is used to construct and test the circuit virtually before hardware implementation. The system is low-cost, easy to understand and suitable for student-level experiments. However, because motion, sensor placement, ambient light and electrical noise can affect the reading, the proposed system should not be used for medical diagnosis.

Keywords: Heart rate, Pulse Sensor, Arduino, BPM, Photoplethysmography, Tinkercad, Biomedical instrumentation

PROJECT GIST

- **Input:** Optical pulse signal from a pulse sensor.
- **Controller:** Arduino Uno.
- **Processing:** Pulse peak detection and BPM calculation.
- **Output:** Heart rate value in beats per minute.
- **Platform:** Tinkercad Circuits simulation.
- **Main purpose:** Educational demonstration of physiological sensing and Arduino programming.

1. INTRODUCTION AND PROJECT CONTEXT

1.1 Background

The human heart continuously pumps blood through the circulatory system. Each cardiac cycle produces changes in blood flow that can be sensed at locations such as the fingertip. Heart rate is normally expressed as beats per minute (BPM). Monitoring heart rate is useful in exercise, health education, physiological experiments and many wearable devices.

1.2 Project Context

Traditional medical monitoring equipment can be complex and expensive. For learning purposes, a microcontroller-based system provides a simple way to understand sensing, signal acquisition, processing and display. Arduino is particularly useful because it provides analog input pins, digital processing and an accessible programming environment.

1.3 Why Tinkercad?

Tinkercad Circuits allows students to construct an electronic circuit virtually, upload Arduino code and observe the simulation without immediately requiring physical components. It reduces wiring mistakes during the learning stage and helps verify the basic logic of the project.

1.4 Problem Statement

To design and simulate a simple, low-cost system that senses a pulse signal and converts it into an understandable heart-rate value using an Arduino.

1.5 Objectives

- To understand the physical principle behind pulse sensing.
- To interface a pulse sensor with Arduino.
- To acquire and process an analog sensor signal.
- To estimate heart rate in BPM.
- To display the measured value in an understandable form.
- To study practical limitations and possible applications.

2. PRINCIPLE OF OPERATION

2.1 Photoplethysmography (PPG)

The pulse sensor works on an optical sensing principle commonly associated with photoplethysmography (PPG). A light source illuminates tissue and a light-sensitive element detects changes in the received light. When the heart pumps blood, the blood volume in the tissue changes periodically. This changes the amount of light detected by the sensor, producing a varying electrical signal related to the pulse.

2.2 BPM Calculation

If the time between two consecutive detected pulse peaks is T seconds, an approximate heart rate can be calculated as:

$$\text{BPM} = 60 / T$$

In a digital implementation, the Arduino can measure the time interval between successive detected peaks using the millisecond timer and convert the interval into BPM. A practical program also uses a threshold and timing condition to avoid counting the same pulse multiple times.

2.3 Main Components and Individual Cost (Indicative)

Component	Quantity	Approx. Unit Cost (₹)	Purpose
Arduino Uno	1	500	Microcontroller and signal processing
Pulse Sensor	1	250	Detects pulse-related optical changes
16× 2 LCD (optional)	1	100	Displays BPM locally
Breadboard	1	80	Temporary circuit assembly
Jumper wires	1 set	80	Electrical connections
USB cable	1	100	Programming/power
Resistor / basic parts	As needed	30	Circuit support

Note: Costs are approximate student-project estimates and may vary by supplier and component quality.

3. CIRCUIT CONNECTION AND INTERFACING

3.1 Pulse Sensor Connection

For a common three-wire pulse sensor module:

Pulse Sensor Pin	Arduino Uno	Function
VCC	5V	Power supply
GND	GND	Common ground
Signal	A0	Analog pulse signal

3.2 LCD Connection (Optional Hardware Extension)

If a standard 16× 2 parallel LCD is used, it can be connected to Arduino digital pins for displaying the BPM. An I2C LCD is another convenient option because it reduces the number of required wires. In Tinkercad, the exact LCD component available should be selected and its pin configuration checked before wiring.

3.3 How the Connections Work

- The pulse sensor receives power from the Arduino.
- The sensor's ground is connected to Arduino ground to provide a common reference.
- The sensor output is connected to analog pin A0.
- Arduino repeatedly samples the signal from A0.
- The program identifies pulse peaks using a threshold and timing logic.
- The calculated BPM is sent to the Serial Monitor and/or LCD.

3.4 Important Wiring Precautions

- Never connect the sensor output directly to a high-voltage source.
- Ensure VCC and GND are not reversed.
- Keep the sensor stable during measurement.
- Use the same ground reference for all modules.
- Check the exact pin labels of the component used in Tinkercad or on the physical module.

4. SOFTWARE DESIGN AND ALGORITHM

4.1 Software Requirements

- Tinkercad Circuits for virtual simulation.
- Arduino-compatible programming environment.
- Arduino Uno board model.
- Pulse sensor model or an equivalent analog pulse-signal source in the simulator.

4.2 Algorithm

1. Start the Arduino and initialize the serial communication.
2. Read the pulse sensor signal from analog pin A0.
3. Compare the signal with a selected threshold.
4. When a valid rising pulse is detected, record the time.
5. Calculate the time interval between consecutive valid peaks.
6. Convert the interval into BPM using $BPM = 60/T$.
7. Display the BPM value.
8. Repeat the process continuously.

4.3 Illustrative Arduino Program

```
const int pulsePin = A0;
int threshold = 550;
bool pulseDetected = false;
unsigned long lastBeat = 0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int signal = analogRead(pulsePin);
  unsigned long now = millis();

  if (signal > threshold && !pulseDetected) {
    if (lastBeat > 0) {
      unsigned long interval = now - lastBeat;
      if (interval > 300 && interval < 2000) {
        int bpm = 60000 / interval;
        Serial.print("Heart Rate: ");
        Serial.print(bpm);
        Serial.println(" BPM");
      }
    }
    lastBeat = now;
    pulseDetected = true;
  }

  if (signal < threshold) {
    pulseDetected = false;
  }

  delay(5);
}
```

The threshold value is only an example. In an actual sensor, the threshold and filtering method should be adjusted according to the signal level and noise. More advanced versions can use filtering, adaptive thresholds and averaging.

5. USES, APPLICATIONS, ADVANTAGES AND LIMITATIONS

5.1 Sensor Uses Separately

Sensor/Part	Separate Use	Role in This Project
Pulse Sensor	Detects pulse-related changes in blood flow	Primary physiological input
Arduino analog input	Measures varying voltage signals	Reads the pulse waveform
LCD (optional)	Displays numerical information	Shows BPM without a computer
Serial Monitor	Displays text/numerical output	Useful during simulation and testing

5.2 Uses of the Project

- Demonstration of biomedical instrumentation principles.
- Learning Arduino sensor interfacing.
- Understanding analog signal acquisition.
- Student laboratory and mini-project demonstrations.
- Basic exercise or educational monitoring prototypes.

5.3 Applications

- Wearable health-monitoring prototypes.
- Fitness and exercise monitoring systems.
- Educational biomedical electronics.
- IoT-based physiological monitoring concepts.
- Embedded-system demonstrations and student projects.

5.4 Advantages

- Low-cost and easy to build.
- Simple Arduino programming.
- Tinkercad enables safe virtual testing.
- Real-time output is possible.
- Can be extended with LCD, Bluetooth, IoT or data logging.

5.5 Disadvantages / Limitations

- Motion can introduce noise and false readings.
- Improper sensor placement can affect the signal.
- Ambient light may influence optical sensors.
- The simple threshold method is less robust than clinical algorithms.
- Component quality and calibration affect accuracy.
- This educational prototype must not be treated as a medical diagnostic device.

6. RESULTS, DISCUSSION AND COST ANALYSIS

6.1 Expected Simulation Result

When the circuit is simulated correctly and a suitable pulse waveform is provided, the Arduino detects periodic peaks and calculates an estimated heart rate. The output is expected to appear as BPM values in the Serial Monitor. A hardware version can additionally display the value on a 16 × 2 LCD.

6.2 Example Observation Table

Trial	Peak Interval (s)	Calculated BPM	Remark
1	1.00	60	Stable example
2	0.86	≈70	Stable example
3	0.75	80	Stable example
4	0.67	≈90	Stable example

These values are illustrative examples for explaining the calculation; they are not measurements from a real person.

6.3 Estimated Total Project Cost

Item	Approx. Cost (₹)	Status
Arduino Uno	500	Required
Pulse Sensor	250	Required
16× 2 LCD	100	Optional
Breadboard	80	Required
Jumper wires	80	Required
USB cable	100	Required
Resistors/basic parts	30	As needed

Estimated basic hardware total: ₹ 1,040. With the optional LCD: approximately ₹ 1,140. Prices are indicative.

6.4 Future Scope

- Add an I2C LCD or OLED display.
- Store heart-rate readings on an SD card.
- Transmit readings through Bluetooth or Wi-Fi.
- Use digital filtering and adaptive peak detection.
- Develop a mobile/web dashboard for data visualization.
- Add additional physiological sensors for a broader monitoring prototype.

7. CONCLUSION AND REFERENCES

7.1 Conclusion

The Heart Rate Monitor Using Arduino project demonstrates how a physiological signal can be sensed, converted into an electrical signal and processed by a microcontroller. The pulse sensor provides the input, Arduino performs the timing and BPM calculation, and a serial monitor or LCD can present the result. The Tinkercad simulation makes the project suitable for learning circuit connections and embedded programming before physical construction.

From a physics perspective, the project connects optical sensing with the periodic changes associated with blood flow. From an electronics perspective, it introduces sensors, analog-to-digital conversion, microcontroller processing and display. Although the prototype is useful for education and experimentation, accurate clinical heart-rate measurement requires validated hardware, appropriate signal processing, calibration and medical-grade testing.

7.2 References / Suggested Reading

- Arduino documentation and Arduino Uno technical resources.
- Autodesk Tinkercad Circuits documentation and tutorials.
- Introductory texts on biomedical instrumentation and physiological measurement.
- Basic references on photoplethysmography (PPG) and optical pulse sensing.
- Component datasheets for the specific pulse sensor and Arduino board used.

7.3 Short Viva Questions

- What is heart rate? — The number of heart beats per minute (BPM).
- What sensor is used? — A pulse sensor based on optical pulse detection.
- Why is Arduino used? — To acquire and process the sensor signal.
- What does BPM mean? — Beats Per Minute.
- Why is Tinkercad useful? — It allows virtual circuit construction and testing.
- What is PPG? — Photoplethysmography, an optical method for detecting blood-volume changes.
- What can cause errors? — Motion, poor placement, noise and ambient light.

END OF PROJECT REPORT